
Alignment between Brains and AI: Evidence for Convergent Evolution across Modalities, Scales and Training Trajectories

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Abstract

Artificial and biological systems may converge on similar computational strategies despite different architectures and learning mechanisms—a form of convergent evolution. We test this at scale by comparing internal representations of 630 AI models (language and vision; 1.33M–72B parameters) against fMRI from the Natural Scenes Dataset, producing over 60 million alignment measurements. Within each modality, higher-performing models spontaneously develop stronger brain correspondence (language: $r = 0.89$; vision: $r = 0.53$); because the inputs are image-evoked, the language results reflect visual-semantic alignment rather than a direct cross-modal comparison. Longitudinal analysis combined with bidirectional Granger tests further shows that past alignment predicts future performance more reliably than the reverse, identifying brain-like representations as a robust early-emerging correlate of learning. Modality-specific organization also emerges: language models align with limbic and integrative regions, vision models with visual cortical hierarchies.

1. Introduction

The rapid progress of Artificial Intelligence (AI) has raised a fundamental question: as these systems increasingly match or surpass human-level performance in language, vision, and reasoning domains (LeCun et al., 2015; Silver et al., 2016; Brown et al., 2020), do their internal representations also converge toward brain-like computational strategies?

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Recent studies have shown that advanced AI systems can now successfully pass Turing tests, with humans unable to reliably distinguish their outputs from those of other humans (Jones & Bergen, 2025). Understanding these internal representational mechanisms is increasingly urgent both for advancing our knowledge of intelligence and for guiding the development of safer, more interpretable AI systems.

Recent advances in Large Language Models (LLMs) (Brown et al., 2020; Grattafiori et al., 2024; Yang et al., 2024) and vision models (Dosovitskiy et al., 2020; Bommasani et al., 2021) provide an unprecedented opportunity to empirically test this convergent evolution hypothesis. Prior research has established correspondence between convolutional neural network activations and hierarchical visual processing in primates, with higher DNN layers correlating with activity in higher visual cortex regions (Yamins et al., 2014; Kriegeskorte, 2015; Horikawa & Kamitani, 2017). Similarly, transformer-based models show representational similarities to human language processing regions (Schrimpf et al., 2021; Caucheteux & King, 2022; Mischler et al., 2024). However, existing work has been constrained by three key limitations: analyses restricted to single modalities rather than cross-modal comparisons, examination of narrow cortical regions rather than distributed networks, and reliance on static model checkpoints that fail to capture representational evolution during training (Storrs et al., 2021; Zhuang et al., 2017).

Convergent evolution—where organisms independently evolve similar traits under similar selective pressures—provides a powerful framework for understanding brain-AI alignment (Conway Morris, 2010; McGhee, 2011). Just as vertebrates and cephalopods independently evolved camera-like eyes under similar visual processing demands (Nilsson, 2013; Ogura et al., 2004), and streamlined body shapes emerged separately in sharks, dolphins, and ichthyosaurs adapting to similar environmental constraints (Werth et al., 2000), artificial and biological systems may converge on similar computational strategies when facing shared information-processing challenges (Zador, 2019; Lillicrap et al., 2020; Hassabis et al., 2017). If demonstrated, such convergence would have profound implications for both understanding biological intelligence and develop-

ing artificial systems. Brain-AI alignment thus serves as both a diagnostic tool and a blueprint for future intelligence architectures.

To systematically test this convergent evolution hypothesis and address the limitations of prior work, we present a comprehensive analysis of brain-AI alignment unprecedented in scale. Our study examines internal representations from over 600 models across diverse architectures, scales, training trajectories, and modalities. Through analyzing 60 million alignment measurements, we directly compare layer-wise activations to human neural recordings, addressing three fundamental questions: (1) Does brain alignment systematically precede performance improvements during training, and is this temporal pattern directional rather than purely coincidental? (2) Do alignment patterns differ systematically between vision and language modalities, or converge toward universal principles? (3) How do model hierarchies progressively map onto cortical processing levels from sensory to associative regions? Our findings reveal that artificial and biological intelligence, despite their distinct evolutionary paths, indeed converge toward similar computational solutions. This establishes fundamental principles that could transform both our understanding of intelligence and the development of future AI systems.

Conflict of Interest Disclosure. All authors are affiliated with academic research institutions (Chinese Academy of Sciences and University of Chinese Academy of Sciences). The AI models and datasets analyzed in this work are publicly released artifacts; none of the authors hold financial interests in, or receive compensation from, the organizations that developed the evaluated models or benchmarks. The authors declare no financial conflicts of interest.

2. Method

2.1. Experimental Framework

We developed a systematic large-scale framework to comprehensively assess brain-AI representational alignment across sensory modalities, model scales, and training dynamics (Figure 1). For brain data, we leveraged fMRI recordings from the Natural Scenes Dataset (NSD) (Allen et al., 2022), which captures neural activity from four subjects viewing thousands of naturalistic images. These images, sourced from the COCO dataset (Lin et al., 2014) and paired with human-generated captions, enabled multimodal analyses across both vision and language domains.

Our AI model collection comprised 630 neural networks spanning diverse architectures and scales: 36 large language models (0.5-72B parameters) from diverse families including Qwen (Yang et al., 2024), Llama (Grattafiori et al., 2024), Gemma (Team et al., 2024), Mistral (Jiang

et al., 2023), Phi (Abdin et al., 2024), and others (Team, 2023; Lambert et al., 2024; Meng et al., 2024; Biderman et al., 2023); and 594 vision models (1.33-1014M parameters) spanning CNNs (Krizhevsky et al., 2012), Transformers (Dosovitskiy et al., 2020), and hybrid architectures (Dai et al., 2021; Wightman, 2019). For performance evaluation, we utilized LLM Leaderboard 2 (Fourrier et al., 2024) with component benchmarks including BBH (Suzgun et al., 2023), MMLU-PRO (Wang et al., 2024), and Chatbot Arena (Chiang et al., 2024) for language models; and ImageNet (Deng et al., 2009) with robustness variants ImageNet-A (Hendrycks et al., 2021b), ImageNet-R (Hendrycks et al., 2021a), and ImageNet-Sketch (Wang et al., 2019) for vision models. In total, our analysis generated over 60 million distinct alignment measurements across different combinations of models, layers, brain regions, subjects, and kernel sizes.

2.2. Centered Kernel Alignment

To quantify representational similarity between AI model activations and brain activity patterns, we employed Centered Kernel Alignment (CKA) (Kornblith et al., 2019), a robust similarity metric invariant to orthogonal transformations. The CKA between two representation matrices X and Y is:

$$\text{CKA}(K, L) = \frac{\text{HSIC}(K, L)}{\sqrt{\text{HSIC}(K, K) \times \text{HSIC}(L, L)}}, \quad (1)$$

where K and L are kernel matrices derived from X and Y , and HSIC is the Hilbert-Schmidt Independence Criterion. We utilized the Radial Basis Function (RBF) kernel (Schölkopf & Smola, 2002) with multiple kernel sizes to capture representational similarities at different scales, from local feature-specific patterns to global representational structures.

Cortical regions were defined using the HCP_MMP1 parcellation (Glasser et al., 2016) (180 regions) and grouped according to the Yeo-7 functional network taxonomy (Yeo et al., 2011), enabling systematic comparison between AI model layers and specific functional brain systems. For longitudinal analyses, we tracked representational evolution in the Pythia language model family (Biderman et al., 2023) (70M-2.8B parameters) and MixNet vision model family (Tan & Le, 2019) (S-XL sizes) throughout training, evaluating alignment at multiple checkpoints across the full training trajectory.

Aggregation. The base tensor is one CKA score per (model, layer, region, subject, kernel) tuple. Each model’s L layers are linearly mapped to 8 normalized depth bins, and subject scores are uniform-averaged across the four NSD subjects (inter-subject consistency $r > 0.97$; Appendix B). The headline “>60M measurements” counts $630 \times \bar{L} \approx 30 \times 180 \times 4 \times 12 \approx 5.95 \times 10^7$ CKA tuples

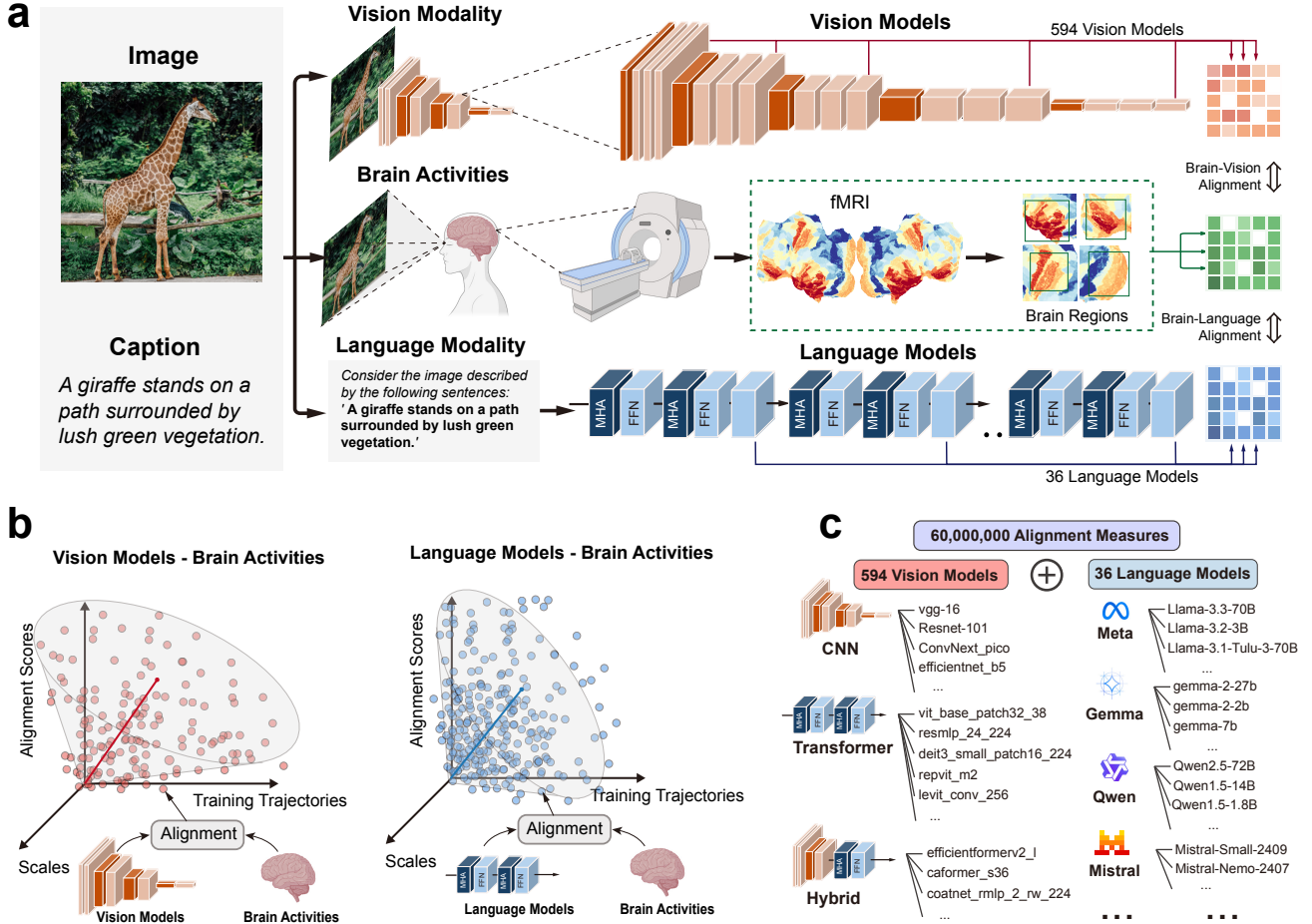


Figure 1. Experimental framework for brain-AI alignment analysis. **a**, Methodological overview showing the multi-modal analysis framework for vision models, brain recordings, and language models. **b**, Multi-dimensional analytical approach spanning scales, training trajectories, and modalities. **c**, Comprehensive analysis of 594 vision models and 36 language models, enabling 60 million distinct alignment measurements.

(KNN-based variants raise the full pool to $\sim 8.5 \times 10^7$). The kernel-scale shift Δ (Section 3.4) is the absolute CKA difference between kernel sizes 68 and 28 per region. Because RBF-CKA operates on $n_{\text{stim}} \times n_{\text{stim}}$ kernel matrices, results are insensitive to post-PCA feature dimensionality (Kornblith et al., 2019).

3. Results

We present our findings organized around four key dimensions of brain-AI alignment: (1) the relationship between model performance and brain alignment across modalities, (2) architectural correspondences between model layers and brain networks, (3) temporal dynamics of alignment during training, and (4) multi-scale alignment patterns across representational granularities. Together, these analyses provide comprehensive evidence for convergent evolution between artificial and biological intelligence systems.

3.1. Performance-Alignment Correlation

Our first key finding is a robust, positive correlation between model performance and brain alignment across both language and vision domains (Figure 2). For language models evaluated on LLM Leaderboard 2 (Fourrier et al., 2024), brain alignment showed a strong correlation with performance (Pearson $r = 0.89$, $p < 7.5 \times 10^{-13}$; Spearman $\rho = 0.88$, $p < 2.7 \times 10^{-12}$), with the relationship better characterized by a logarithmic fit ($R^2 = 0.80$) than a linear one ($R^2 = 0.78$). This pattern held across different model families.

The logarithmic relationship suggests diminishing returns: initial performance improvements correspond to substantial alignment gains, followed by more modest increases. Notably, models with similar performance but different fine-tuning approaches showed distinct alignment scores. For instance, Llama-3.1-Tulu-3-8B (Lambert et al., 2024) exhibits higher alignment than base Llama-3.1-8B (Grattafiori

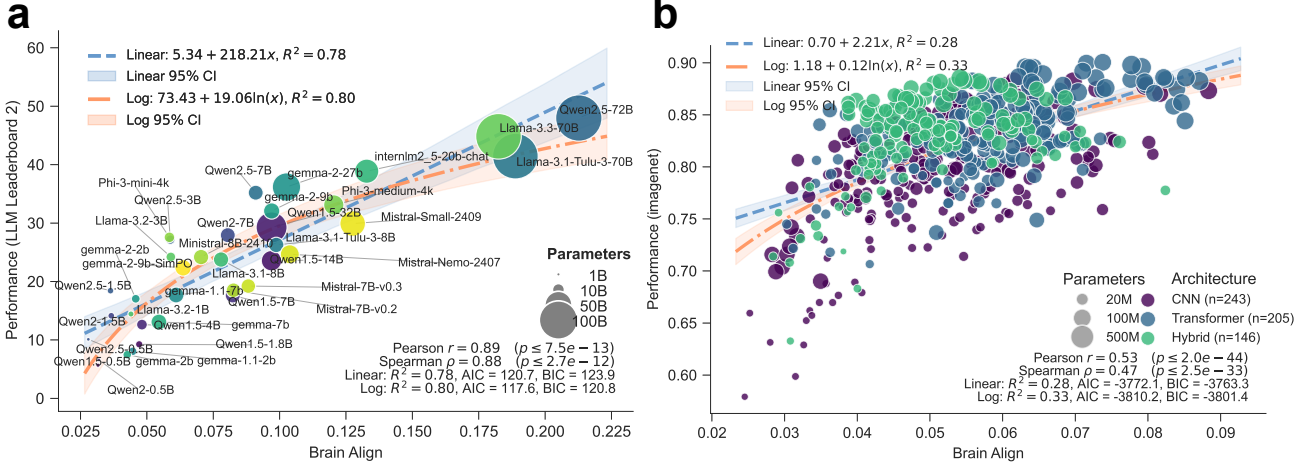


Figure 2. Relationship between model performance and brain alignment. **a**, Language models show strong correlation between LLM Leaderboard 2 (Fourrier et al., 2024) performance and brain alignment scores, with logarithmic fit outperforming linear fit. Parameter counts indicated by marker size. **b**, Vision models demonstrate moderate but significant correlation between ImageNet top-1 accuracy and brain alignment, with differentiation between architectural types.

Table 1. Correlation between model performance and brain alignment across benchmarks. FDR-corrected p-values account for multiple comparisons.

Model	Benchmark	r	p-value	ρ	p-value
Language	BBH	0.84	<2.3e-10	0.86	<1.9e-11
	MMLU-PRO	0.83	<3.8e-10	0.85	<7.3e-11
	Leaderboard 2	0.89	<7.5e-13	0.88	<2.7e-12
	Chatbot Arena	0.73	<2.3e-05	0.80	<9.9e-07
Vision	ImageNet	0.53	<2.0e-44	0.47	<2.5e-33
	ImageNet-A	0.46	<7.5e-32	0.45	<2.1e-31
	ImageNet-R	0.52	<8.3e-43	0.47	<1.1e-33
	sketch	0.53	<1.3e-43	0.47	<4.0e-34

et al., 2024), while gemma-2-9b-SimPO (Meng et al., 2024) shows lower alignment than standard gemma-2-9b (Team et al., 2024), indicating that training methodologies impact biological alignment.

For vision models evaluated on ImageNet (Deng et al., 2009), we observed a significant correlation between accuracy and brain alignment (Pearson $r = 0.53$, $p < 2.0 \times 10^{-44}$; Spearman $\rho = 0.47$, $p < 2.5 \times 10^{-33}$). The correlation was consistent across model sizes, indicating a scale-invariant relationship. Transformer-based vision models exhibited slightly higher brain alignment than CNNs at equivalent performance levels.

Both vision and language models exhibit a logarithmic relationship between performance and brain alignment. Table 1 presents correlation statistics across multiple benchmarks. The consistent brain-performance correlation across vastly different architectures provides compelling evidence for convergent evolution.

3.2. Architectural Correspondences

We examined how brain alignment varies across layers of deep neural networks (Figure 3a). Vision models exhibited a gradual increase in alignment with depth, peaking at the final segments (6/8 to 8/8), mirroring the hierarchical organization of the visual cortex from concrete feature detection to abstract object recognition. In contrast, language models showed overall higher brain alignment, with a distinct middle-layer peak (3/8 to 5/8) that remained relatively stable through deeper layers, suggesting they engage with more abstract, integrated representations similar to those in higher cognitive networks.

When examining alignment with specific brain networks (Figure 3b), we observed clear modality-specific patterns. At shallow layers (2/8), vision models showed modest alignment with a slight preference for visual regions, while language models demonstrated stronger alignment with limbic and default mode networks—regions associated with higher-level cognitive integration rather than primary sensory processing. At deeper layers (6/8), vision models developed substantially stronger alignment with visual network regions, while language models maintained strong alignment to limbic and default regions across networks. Notably, language models showed strongest correspondence to limbic, default mode, and frontoparietal networks, which are associated with semantic memory, self-referential processing, and executive functions respectively (Goldstein et al., 2022; Dobs et al., 2022).

Dimensionality reduction analysis (Figure 3c) revealed two distinct manifolds for vision and language model representations. Notably, deeper vision model layers positioned closer to the language model manifold, suggesting that as

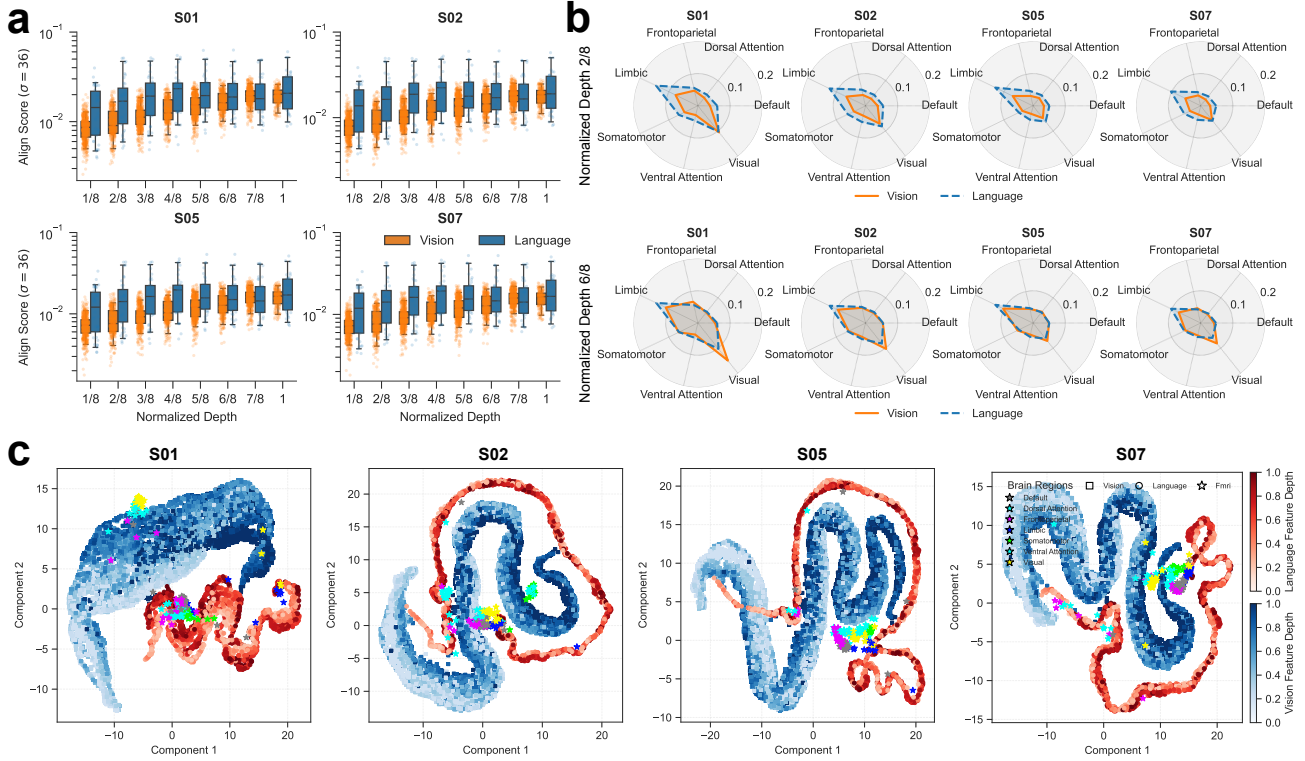


Figure 3. Layer-wise brain alignment patterns. **a**, Brain alignment by normalized layer depth for vision and language models. Each model’s layers were normalized into 8 depth segments. **b**, Comparison of brain network alignment at normalized depths 2/8 and 6/8 across Yeo-7 brain networks (Yeo et al., 2011). **c**, UMAP (McInnes et al., 2018) visualization showing relationships between model layers and brain regions.

visual processing becomes more abstract, it shares representational properties with language processing (Binder et al., 2016; Huth et al., 2016). Quantifying this trend, the Wasserstein distance between vision- and language-model alignment profiles decreases monotonically across depth bins (from 0.0152 at depth 1/8 to 0.0056 at depth 8/8, Spearman $\rho = -1.00$; cosine similarity rises from 0.983 to 0.995, $\rho = +1.00$; Appendix G). Brain regions followed a parallel pattern: visual cortex regions clustered near early vision model layers, while association cortices positioned closer to language model representations.

3.3. Brain Alignment as an Early-Emerging Correlate of Learning

Understanding whether brain alignment precedes performance improvements is crucial for distinguishing between performance-driven convergence and alignment-driven performance. Longitudinal analysis revealed that brain alignment is not merely a static property but evolves dynamically throughout training (Figure 4a-d).

A critical finding emerged: increases in brain alignment consistently preceded improvements in task performance. For vision models, brain alignment increased throughout

training, reaching approximately 85% of maximum alignment by just 20% of training samples, while performance only began improving after approximately 5% of training. The correlation between brain alignment and performance was strong ($r = 0.98$ for MixNet-L). This temporal precedence indicates that brain-like representations emerge as an early correlate of effective learning rather than merely a by-product of late-stage optimization (Saxe et al., 2019; Kriegeskorte, 2015).

This leading indicator relationship was even more pronounced in language models, where brain alignment began increasing as early as 1% of training and reached stable levels by approximately 10% of training tokens. Performance metrics showed significant delays, only reaching half of their maximum values after approximately 10% of training. Across all Pythia models, correlations ranged from $r = 0.73$ to $r = 0.95$, with brain alignment increases consistently preceding performance improvements. This pattern held across model scales and modalities. To move beyond temporal precedence and assess directionality, we conducted bidirectional Granger causality tests on all longitudinal trajectories (Section 3.5); the asymmetric directional pattern further supports the convergent evolution hypothesis without requiring an interventional manipulation (Zhuang et al., 2021;

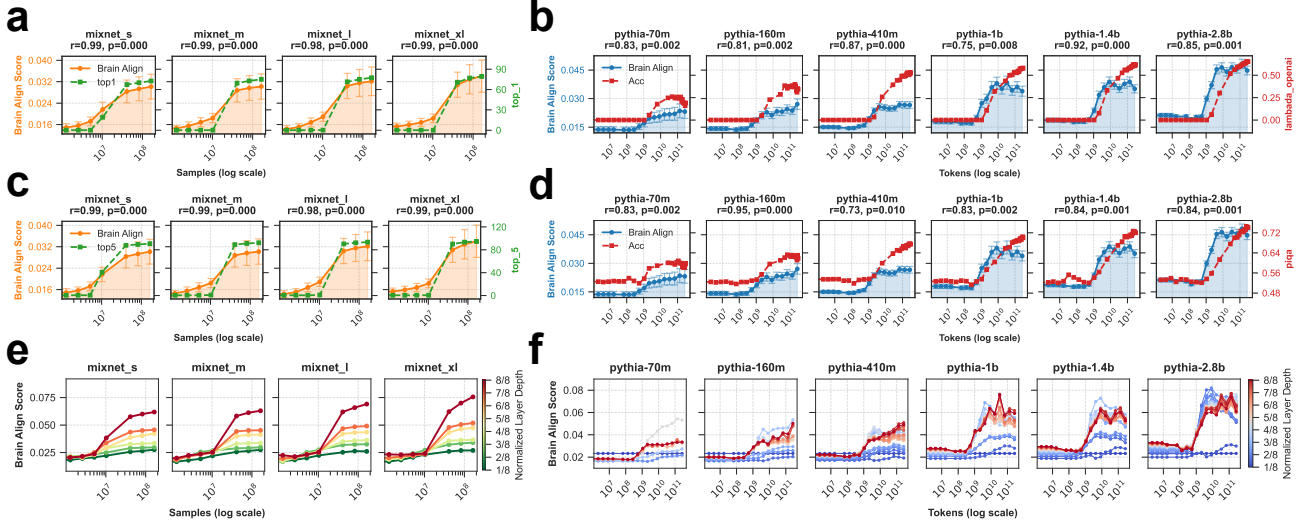


Figure 4. Evolution of brain alignment during training. **a,c**, Dual y-axis plots for the MixNet (Tan & Le, 2019) vision family showing brain alignment (left axis) against ImageNet top-1 (**a**) and top-5 (**c**) accuracy (right axis) across training. **b,d**, Same layout for the Pythia (Biderman et al., 2023) language family, with brain alignment plotted against LAMBADA (**b**) and PIQA (**d**) accuracy. **e**, Layer-wise brain alignment evolution for MixNet across training progression; **f**, same for Pythia.

Lillicrap et al., 2020).

Layer-wise analysis revealed distinct modality-specific patterns (Figure 4e-f). In vision models, brain alignment steadily increased across all layers, with deeper layers achieving higher alignment scores—consistent with the gradual refinement of hierarchical visual representations. Notably, alignment at the final training stages showed a slight decrease, potentially reflecting overfitting to task-specific features that diverge from brain-like representations. In language models, the layer-wise dynamics were more complex: early layers maintained low alignment throughout training, likely because these layers encode low-level token features that do not strongly correspond to the visual-semantic nature of our stimuli. Middle layers showed a characteristic rise-then-fall pattern, suggesting transient computational scaffolding that is subsequently refined. Deeper layers exhibited delayed growth that continued after middle layers declined, indicating that higher-level semantic representations develop later in training. These dynamics became more pronounced in larger models (e.g., Pythia-2.8B vs. Pythia-70M), suggesting that scale amplifies the development of brain-like representational structures and enables more sophisticated hierarchical organization.

3.4. Multi-scale Alignment Patterns

We systematically varied the kernel size in our CKA measurements to explore how measurement scale impacts alignment (Figure 5a). The kernel size parameter in RBF-based CKA controls the sensitivity to representational similarity: smaller kernel sizes capture highly specific representational similarities, analogous to how primary sensory areas re-

spond selectively to specific features, while larger kernel sizes detect similarities across more varied stimuli, reflecting broader representational patterns similar to association cortices (Kornblith et al., 2019). This multi-scale approach enabled detection of alignment patterns that might be specific to particular representational granularities.

As kernel size increased from 28 to 68, primary visual areas showed substantial decreases in relative alignment, while limbic, ventral attention, and default mode networks showed increasing alignment. The most dramatic decreases occurred in early visual areas V1 ($\Delta = -5.30$), V2 ($\Delta = -2.81$), and V3 ($\Delta = -1.05$), consistent with these regions’ specialization for local, feature-specific processing. In contrast, the strongest increases occurred in limbic regions including orbitofrontal cortex (OFC; $\Delta = +1.56$), temporal regions TF ($\Delta = +1.36$), and parahippocampal regions ($\Delta = +1.09$)—areas associated with higher-level integration and contextual processing. These shifts were remarkably consistent across subjects (sign consistency = 0.83), confirming the robustness of the observed scale-dependent regional preferences across individuals.

Cortical surface visualization (Figure 5b) revealed a clear posterior-to-anterior gradient in alignment patterns at different kernel sizes. At smaller kernel sizes, posterior sensory regions showed relatively stronger alignment, while at larger kernel sizes, anterior association areas dominated. This gradient was particularly pronounced for language models in frontal and temporal areas, consistent with these regions’ role in abstract semantic processing (Fedorenko & Varley, 2016; Hagoort, 2019). The systematic shift from posterior sensory regions to anterior association areas as kernel size increased reflects a fundamental organizational principle:

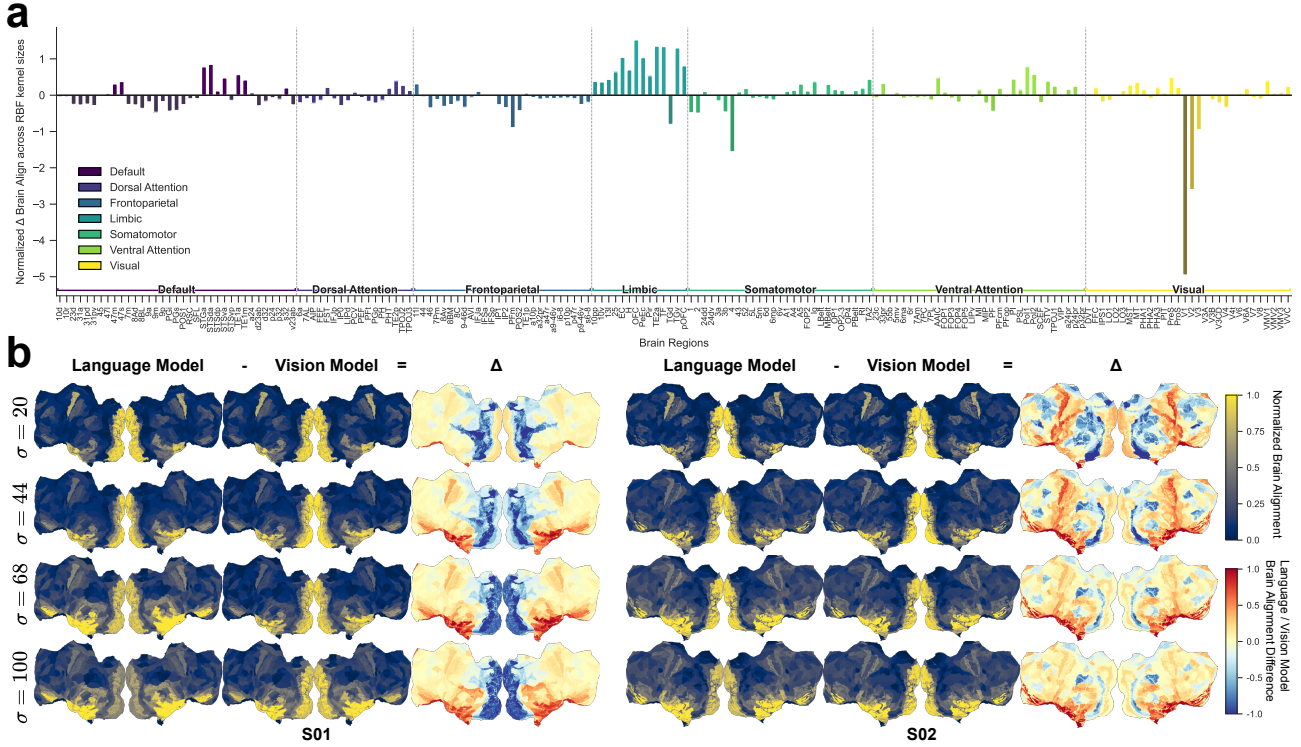


Figure 5. Brain alignment patterns across representational scales. **a**, Normalized brain alignment changes across different brain regions as kernel size increases from 28 to 68. **b**, Cortical surface visualizations showing alignment patterns at four different kernel sizes for language models (left), vision models (middle), and their difference (right, Δ).

as information flows through processing pathways in both biological and artificial systems, representations transition from local, feature-specific encodings to distributed, context-dependent structures (Margulies et al., 2016; Huntenburg et al., 2018). The stronger scale-dependent modulation observed in language models further supports the hypothesis that language processing operates at inherently broader representational scales than visual processing.

Finally, we examined how different models relate to each other when characterized by their brain alignment profiles. Using hierarchical clustering on alignment patterns across 180 cortical regions, 8 depth levels, and 4 subjects, we found that architecture emerges as the dominant organizing principle, with clear separation between CNN-based and transformer-based models (see Appendix F for details). Within these architectural divisions, modality-specific clustering emerged: language models formed several well-defined subclusters corresponding to model families (e.g., LLaMA, Qwen), while vision models clustered by architectural generation (early CNNs vs. modern residual networks). This alignment-driven taxonomy offers a biologically grounded organizational view of the model landscape, complementing traditional task benchmarks by revealing deeper commonalities and distinctions in representational structure that may not be apparent from performance metrics

alone.

3.5. Robustness Analyses

To address concerns about causal interpretation, scale confounds, and absolute alignment magnitudes, we report three complementary controls (Table 2; per-model details in Appendix G). *Bidirectional Granger tests* (Granger, 1969) on each longitudinal trajectory show that alignment $\rightarrow$ performance is significant in 9/10 cases (Pythia 6/6, MixNet 3/4) while the reverse is significant in only 3/10—an asymmetry that rules out shared-trend co-variation as the sole explanation, although Granger quantifies directional predictivity rather than mechanism (Firestone, 2020). *Partial Pearson correlations* controlling for $\log_{10}$ parameter count remain highly significant in both modalities (language $r_{\text{partial}} = 0.54$, $p = 8.8 \times 10^{-4}$; vision $r_{\text{partial}} = 0.31$, $p = 5.1 \times 10^{-14}$), so the performance–alignment trend is not a scale artifact. *Inter-subject CKA noise ceilings* per region show that mean alignment meets or exceeds the ceiling in 78% (language) and 53% (vision) of the 180 HCP regions, with a clean monotonic gradient across benchmark tertiles (Table 2), confirming that top models approach the upper bound recoverable from this dataset.

Table 2. Summary of robustness analyses. Granger results tally lag-1 significant trajectories; partial r controls for $\log_{10}$ parameter count; ceiling shows the percentage of 180 HCP regions whose mean model alignment meets or exceeds the inter-subject noise ceiling.

	Language	Vision
Granger align→perf (sig.)	6/6	3/4
Granger perf→align (sig.)	1/6	2/4
Raw / partial r (vs. params)	0.88 / 0.54	0.50 / 0.31
% regions reaching ceiling (all)	78%	53%
% reaching ceiling (top 1/3)	88%	62%
% reaching ceiling (bottom 1/3)	6%	36%

4. Discussion

Our findings provide compelling evidence for convergent evolution between artificial and biological intelligence systems. Despite fundamentally different origins, architectures, and learning mechanisms, AI models spontaneously develop representations that progressively align with human brain activity patterns as they optimize for task performance (Hasson et al., 2020; Richards et al., 2019; Zador, 2019).

Relation to Prior Convergence Work. The Platonic Representation Hypothesis (Huh et al., 2024) and the Contravariance Principle (Cao & Yamins, 2024) argue that high-performing models converge toward a shared statistical model of reality. Our study is the empirical complement: we replace the model-to-model convergence target with the *human brain* as an external biological anchor, at substantially larger scale (630 models, >60M measurements). Among brain-alignment studies, Conwell et al. (2024) examined 224 vision models on NSD object-selective cortex; we extend to whole-cortex alignment, language models, and training dynamics. Gokce & Schrimpf (2025) and AlKhamissi et al. (2025) reported architecture-/objective-dependent emergence in primate vision and language-evoked LLM trajectories respectively; the latter’s “outgrowth” of the human language network is consistent with the diminishing-returns regime we observe in the largest Pythia checkpoints (Section 3.5).

Performance Alignment and Convergent Evolution. The robust correlation between model performance and brain alignment holds independently within each modality, representing converging evidence for convergent evolution. The logarithmic relationship observed in both modalities suggests a principled trajectory of diminishing returns, where initial improvements yield substantial alignment gains followed by more modest increases. This pattern aligns with theoretical perspectives suggesting that as complex systems approach optimal solutions for information processing challenges, they naturally converge toward similar computational strategies despite different evolutionary pathways (Yamins et al., 2014; Marblestone et al., 2016). We deliberately avoid framing the difference between the two

coefficients ($r = 0.89$ for language vs. $r = 0.53$ for vision) as a direct cross-modal comparison: the language-model inputs are captions paired with image-evoked fMRI, so the corresponding correlations reflect visual-semantic alignment grounded in the NSD paradigm rather than general linguistic competence (Fedorenko & Thompson-Schill, 2014; Pereira et al., 2018). That purely text-trained LLMs nonetheless track image-evoked brain responses is itself a non-trivial signature of cross-modal representational convergence (Schrimpf et al., 2021; Caucheteux & King, 2022).

Hierarchical and Multi-scale Organization. Layer-wise and multi-scale analyses revealed systematic gradients of representational organization that recapitulate aspects of the brain’s processing hierarchy. Vision models show gradual alignment increases with depth, paralleling hierarchical progression from low-level feature detection to high-level object recognition in visual cortex (Yamins & DiCarlo, 2016; Cichy et al., 2016). Language models exhibit stronger, more distributed alignment corresponding to association areas involved in abstract semantic processing (Goldstein et al., 2022). The observation that deeper vision model layers begin to share representational properties with language processing aligns with theories of embodied cognition suggesting visual and language systems converge in higher-level semantic processing (Binder et al., 2016; Huth et al., 2016). Multi-scale analysis revealed a posterior-to-anterior shift consistent with transition from localized, feature-specific representations to distributed, integrative ones (Margulies et al., 2016; Huntenburg et al., 2018).

Temporal Dynamics and Architectural Biases. Longitudinal analysis revealed that brain alignment consistently emerges earlier than measurable performance gains, and bidirectional Granger tests show that past alignment significantly predicts future performance far more reliably than the reverse direction (Saxe et al., 2019; Kriegeskorte, 2015). This phenomenon was observed across architectures and scales, reinforcing the idea that convergent evolution arises as models optimize for performance. The temporal precedence of alignment over performance is particularly striking: in both vision and language models, brain alignment began increasing early in training and reached substantial levels before task performance showed significant improvement. While these analyses establish directional predictivity rather than mechanistic causation (Firestone, 2020), they identify brain-like representations as a robust early-emerging correlate of learning—an observation with implications for both understanding biological intelligence and developing more capable AI systems (Zhuang et al., 2021; Lillicrap et al., 2020).

Architectural Inductive Biases and Model Taxonomy. Our taxonomic analysis based on brain alignment profiles revealed that architecture serves as the dominant organizing

principle, with pronounced separation between CNN-based and transformer-based models. The emergence of modality-specific clustering within these architectural divisions indicates that both structural constraints and task demands shape representational alignment (Kriegeskorte & Douglas, 2018; Storrs et al., 2021). Models with residual connections and attention mechanisms exhibited higher brain alignment scores, suggesting these architectural elements may facilitate greater biological plausibility (Kietzmann et al., 2019; Kubilius et al., 2019). The tight grouping of modern CNNs with residual structures compared to earlier feedforward architectures suggests that skip connections impose a stronger inductive bias toward biologically plausible representations. Similarly, the diverse distribution of hybrid architectures across the taxonomic space reflects their architectural flexibility in balancing convolutional and attentional components.

Limitations. Static fMRI has limited temporal resolution; EEG/MEG would complement our analyses (Cichy et al., 2016). The NSD paradigm captures passive image viewing, and we pair the same image-evoked responses with COCO captions, so our “language model–brain” results probe visually grounded semantic alignment rather than general linguistic competence (Allen et al., 2022; Firestone, 2020); cross-modal observations are therefore descriptive. Our similarity measure is a single family (RBF-CKA), and brain-alignment conclusions can shift across metrics (Williams et al., 2021; Gröger et al., 2026); cross-kernel and KNN-based checks (Appendix C) mitigate but do not exhaust this concern. Prior work also shows benchmark-optimized models can *worsen* as IT models (Linsley et al., 2023) and that behavioral parity need not imply shared competence (Suresh et al., 2023); our ceiling analysis (Section 3.5) shows top models sit close to, not below, the ceiling, but this calibration is NSD-specific. Finally, benchmark contamination may inflate performance estimates for the largest LLMs (Dodge et al., 2021; Schaeffer et al., 2023).

Implications. These results have significant implications for both neuroscience and AI. For neuroscience, the spontaneous emergence of brain-like representations in optimized AI systems provides computational evidence that the brain’s organizational principles may reflect optimal solutions to information processing challenges rather than evolutionary accidents (Kriegeskorte & Douglas, 2019; Richards et al., 2019). The distinctive alignment patterns across vision and language modalities offer computational perspectives on the functional specialization observed across cortical regions (Kahn et al., 2010; Binder et al., 2016). The systematic gradient from posterior sensory to anterior association regions observed in our multi-scale analysis mirrors established understanding of cortical organization, where earlier sensory regions process concrete features while higher association areas integrate information at broader conceptual

scales (McIntosh, 2000; Mesulam, 1998).

For AI development, our findings suggest that incorporating explicit brain-inspired constraints could potentially accelerate model development by guiding optimization toward more effective representations (Hassabis et al., 2017; Lindsay, 2021). The observation that brain alignment emerges as an early-stage correlate of learning particularly suggests that neuroscientific insights could help identify promising architectural innovations or training regimes (Lindsay, 2021; Richards et al., 2019). Furthermore, the alignment-driven taxonomy we developed provides a novel framework for comparing models that complements traditional task-based benchmarks, potentially revealing representational similarities and differences not captured by performance metrics alone. Understanding how and why alignment emerges may ultimately bridge the gap between natural and artificial intelligence, enabling the development of AI systems that are not only more capable but also more interpretable and aligned with human cognition.

Impact Statement

This paper presents work whose goal is to advance the field of Machine Learning by establishing connections between artificial and biological intelligence systems. Understanding brain-AI alignment has potential implications for developing more interpretable AI systems and for neuroscience research. We believe this research contributes positively to both fields. There are many potential societal consequences of our work, none which we feel must be specifically highlighted here.

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A. Additional Methodological Details

A.1. Data Acquisition and Preprocessing

fMRI Data. We utilized the Natural Scenes Dataset (NSD) (Allen et al., 2022), a large-scale fMRI dataset capturing neural responses to natural images. From the eight subjects in the dataset, we selected four (S01, S02, S05, S07) who had complete data trajectories. Each subject’s data included responses to 24,980 training stimuli (unique per subject) and 2,770 testing stimuli (shared across subjects). We extracted functional responses from the preprocessed `func1pt8mm` data following the standard NSD preprocessing pipeline, which includes motion correction, distortion correction, and hemodynamic response modeling.

Stimulus Materials. The NSD stimuli consist of images from the COCO dataset (Lin et al., 2014), which provides paired image-caption annotations. For each image, we extracted the first available caption as textual input for language models, enabling aligned multimodal analyses.

A.2. Neural Network Models

Language Models. We analyzed 36 contemporary language models spanning 500M to 72B parameters, including Qwen (Yang et al., 2024), Llama (Grattafiori et al., 2024), Gemma (Team et al., 2024), Mistral (Jiang et al., 2023), Phi (Abdin et al., 2024), and specialized variants such as SimPO (Meng et al., 2024) and Tulu (Lambert et al., 2024). All models were implemented using the HuggingFace Transformers library (Wolf et al., 2020). We extracted hidden state representations from each transformer block, focusing on the last token representation (Devlin et al., 2019; Liu et al., 2019).

Vision Models. We analyzed 594 vision models spanning CNNs, Transformers, and Hybrids, sourced from the TIMM library (Wightman, 2019) with parameters ranging from 1.33M to 1.014B. We implemented an automated feature extraction pipeline to capture hidden states from each network block.

A.3. CKA Implementation Details

For computational efficiency with high-dimensional representations exceeding 4096 dimensions, we applied PCA to reduce dimensionality while preserving at least 95% of variance. We performed analyses across multiple kernel sizes (20, 28, 36, 44, 52, 60, 68, 72, 84, 92, 100) to capture representational similarities at different scales. All CKA computations were implemented in PyTorch and executed on GPUs with parallelization across models and brain regions.

For brain-region specific analyses, we calculated alignment scores by taking the average of the top 5% highest-aligning brain regions at the normalized model depth of 6/8, providing a sensitive measure of alignment.

A.4. Longitudinal Training Analysis

For language models, we analyzed the Pythia family (Biderman et al., 2023) (70M-2.8B parameters) with checkpoints throughout training. For vision models, we trained MixNet variants (S/M/L/XL) (Tan & Le, 2019) from scratch on ImageNet-1K using standard TIMM configurations, with checkpoints saved at logarithmically spaced intervals.

Performance was tracked using LAMBADA (Paperno et al., 2016) and PIQA (Bisk et al., 2020) for language models, and ImageNet top-1/top-5 accuracy for vision models.

B. Inter-subject Consistency Analysis

Table 3. Inter-subject consistency for language models.

	S01	S02	S05	S07
S01	1.000	0.997	0.999	0.997
S02	0.997	1.000	0.999	1.000
S05	0.999	0.999	1.000	0.999
S07	0.997	1.000	0.999	1.000

Table 4. Inter-subject consistency for vision models.

	S01	S02	S05	S07
S01	1.000	0.970	0.981	0.970
S02	0.970	1.000	0.998	0.999
S05	0.980	0.998	1.000	0.998
S07	0.970	0.999	0.998	1.00

To ensure robustness, we calculated pairwise Pearson correlations between brain-performance relationships across all four

subjects (Tables 3 and 4). For language models, inter-subject correlations ranged from 0.997 to 1.000. For vision models, correlations ranged from 0.970 to 0.999. These high cross-subject correlations indicate that observed representational correspondences reflect universal computational principles rather than individual variations.

C. Metric Consistency Analysis

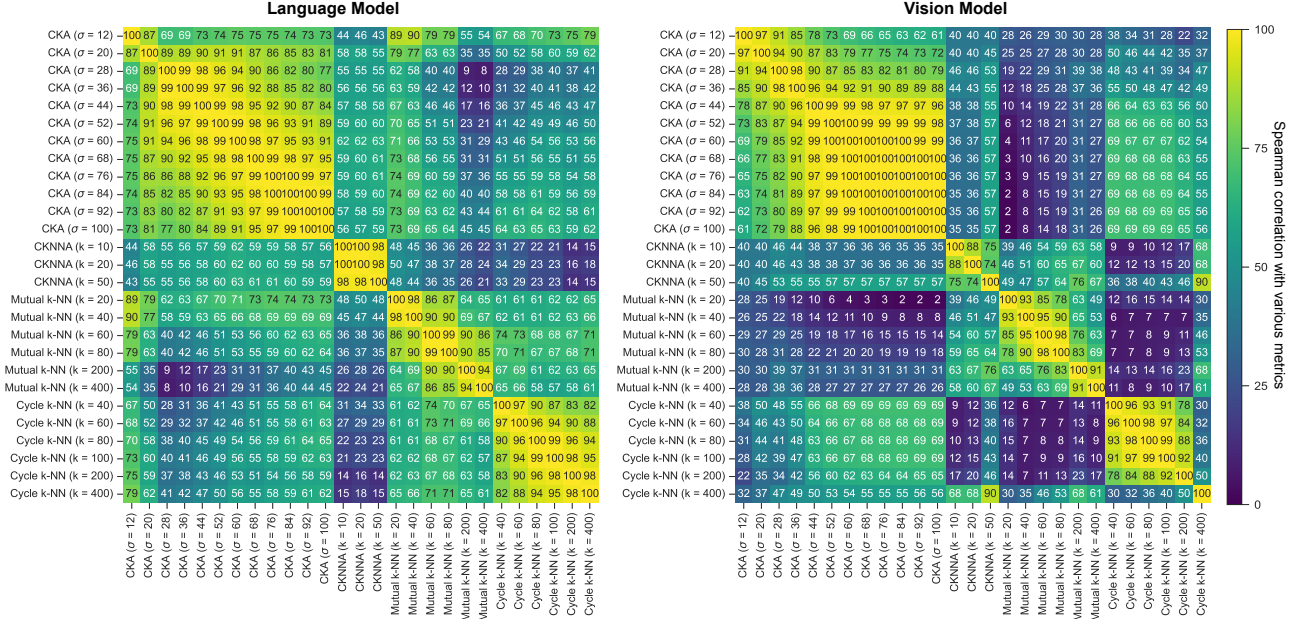


Figure 6. Consistency analysis across different similarity metrics and parameter settings. Heatmaps showing Spearman rank correlations between different similarity measurement approaches for language models and vision models, comparing CKA with varying kernel sizes and k-nearest neighbor (KNN) variants.

To validate the stability of our similarity metrics, we conducted comprehensive consistency analysis across different kernel sizes and alternative KNN-based similarity measures (Figure 6). Using Spearman rank correlation, we confirmed strong rank consistency across different parameter settings, supporting the reliability of our representational similarity framework regardless of specific kernel size choices.

D. Extended Correlation Statistics

Table 5 presents complete correlation statistics between brain alignment and performance across all benchmarks analyzed.

E. Statistical Analysis Details

Correlation Analysis. We assessed performance-alignment relationships using both Pearson and Spearman correlation coefficients. For each correlation, we calculated 95% confidence intervals via bootstrap sampling with 1,000 iterations. Multiple comparisons were corrected using FDR correction.

Regression Models. To characterize performance-alignment relationships, we fitted both linear and logarithmic regression models. Model selection was based on AIC, BIC, and R^2 values. For language models, a logarithmic model ($R^2 = 0.80$, AIC = 117.6, BIC = 120.8) outperformed a linear model ($R^2 = 0.78$, AIC = 120.7, BIC = 123.9). For vision models, a logarithmic fit ($R^2 = 0.33$, AIC = -3810.2, BIC = -3801.4) provided better characterization than a linear model ($R^2 = 0.28$, AIC = -3772.1, BIC = -3763.3).

Model Taxonomy Construction. We constructed hierarchical organization of models based on brain alignment profiles using agglomerative clustering with Ward’s linkage method (Ward Jr, 1963). For each model, we created a feature vector comprising alignment scores across all 180 brain regions, 8 normalized depth levels, and 4 subjects, resulting in a $180 \times 8 \times 4$

Table 5. Complete correlation statistics across all benchmarks.

Model Information		Pearson Correlation				Spearman Correlation			
Model	Benchmark	r	95% CI	p	FDR p	ρ	95% CI	p	FDR p
Language Models	IFEval	0.77	[0.59, 0.88]	<3.5e-08	<4.1e-08	0.79	[0.58, 1.00]	<1.3e-08	<1.6e-08
	BBH	0.84	[0.70, 0.91]	<2.3e-10	<3.0e-10	0.86	[0.65, 1.07]	<1.9e-11	<2.7e-11
	MATH Lvl 5	0.76	[0.58, 0.87]	<7.6e-08	<8.7e-08	0.67	[0.46, 0.88]	<7.1e-06	<7.6e-06
	GPQA	0.76	[0.57, 0.87]	<7.7e-08	<8.7e-08	0.78	[0.57, 0.99]	<2.3e-08	<2.8e-08
	MUSR	0.66	[0.42, 0.81]	<1.3e-05	<1.3e-05	0.59	[0.38, 0.80]	<1.6e-04	<1.6e-04
	MMLU-PRO	0.83	[0.69, 0.91]	<3.8e-10	<5.0e-10	0.85	[0.64, 1.06]	<7.3e-11	<1.0e-10
	Leaderboard 2	0.89	[0.79, 0.94]	<7.5e-13	<1.1e-12	0.88	[0.67, 1.09]	<2.7e-12	<4.0e-12
	Chatbot Arena	0.73	[0.48, 0.87]	<2.3e-05	<2.4e-05	0.80	[0.55, 1.05]	<9.9e-07	<1.1e-06
Vision Models	ImageNet	0.53	[0.47, 0.59]	<2.0e-44	<1.5e-43	0.47	[0.41, 0.52]	<2.5e-33	<6.5e-33
	ImageNet-A	0.46	[0.39, 0.52]	<7.5e-32	<1.7e-31	0.45	[0.40, 0.50]	<2.1e-31	<4.4e-31
	ImageNet-A-Clean	0.51	[0.45, 0.57]	<1.7e-41	<7.0e-41	0.46	[0.41, 0.51]	<2.3e-32	<5.6e-32
	ImageNet-R	0.52	[0.46, 0.58]	<8.3e-43	<4.6e-42	0.47	[0.42, 0.52]	<1.1e-33	<3.1e-33
	ImageNet-R-Clean	0.51	[0.45, 0.57]	<2.8e-40	<1.0e-39	0.45	[0.40, 0.50]	<5.5e-31	<1.1e-30
	ImageNet-Real	0.50	[0.44, 0.56]	<1.2e-38	<4.0e-38	0.44	[0.39, 0.49]	<2.9e-29	<5.0e-29
	ImageNetv2-matched-freq. sketch	0.52	[0.46, 0.57]	<5.6e-42	<2.4e-41	0.45	[0.40, 0.50]	<1.3e-30	<2.4e-30
		0.53	[0.47, 0.58]	<1.3e-43	<8.2e-43	0.47	[0.42, 0.52]	<4.0e-34	<1.2e-33

dimensional representation.

F. Model Taxonomy via Brain Alignment

Using brain alignment patterns as high-dimensional feature vectors ($180 \text{ regions} \times 8 \text{ depth levels} \times 4 \text{ subjects}$), we constructed a hierarchical clustering tree to visualize representational similarity (Figure 7). Architecture emerges as the dominant organizing principle in the dendrogram.

CNNs form two distinct clusters that reflect a generational and structural divide. One smaller cluster contains earlier CNN architectures such as VGG (Simonyan & Zisserman, 2014) and Xception (Chollet, 2017), which lack residual connections and rely on straightforward feedforward stacking of convolutional layers. In contrast, a larger and more internally coherent cluster includes modern CNNs like ResNet (He et al., 2016), MobileNet (Koonce & Koonce, 2021), and EfficientNet (Koonce, 2021), which employ residual or shortcut connections. These architectural innovations appear to yield more brain-aligned representations. The tight grouping of modern CNNs suggests that residual structures impose a stronger inductive bias toward biologically plausible representations.

Transformer-based models occupy a broader and more internally diverse region of the tree. Within this space, language models form several well-defined subclusters that often correspond to model families such as LLaMA (Grattafiori et al., 2024) or Qwen (Yang et al., 2024), suggesting that architectural lineage and fine-tuning methodology significantly shape alignment patterns. Language models tend to cluster separately from vision models, even among transformers, indicating a modality-driven divergence in representational geometry at the model level.

Hybrid architectures, which integrate convolutional and attention-based components (e.g., ConvNeXt (Liu et al., 2022), CoAtNet (Dai et al., 2021), FocalNet (Yang et al., 2022)), display a more scattered distribution across the dendrogram. This dispersion reflects their architectural flexibility and suggests that hybrid models do not converge toward a single representational style, but instead form a continuum of brain alignment profiles based on their relative balance of convolutional and attentional components.

To verify that the brain-based taxonomy is not a redundant view of benchmark performance, we conducted a Mantel test (Mantel, 1967) comparing two pairwise model-distance matrices: (1) Euclidean distance in the brain-alignment vector ($180 \times 8 \times 4$), and (2) Euclidean distance in the benchmark performance vector (LLM Leaderboard 2 subscores for language; ImageNet and robustness variants for vision). The two distance structures are essentially uncorrelated (vision: Mantel $r = 0.020$, $p = 0.095$, $n = 594$; language: Mantel $r = -0.061$, $p = 0.534$, $n = 38$), demonstrating that the alignment-based taxonomy carries information that benchmark performance alone cannot recover. For example, Tulu-3-8B and Gemma-2-27b-it have near-identical performance profiles (cosine distance 0.0005) yet highly divergent brain-alignment profiles (cosine distance 0.76).

G. Robustness Details

This appendix expands on the three robustness analyses summarized in Section 3.5.

Bidirectional Granger causality. For each longitudinal trajectory we fit a bivariate vector-autoregressive model at lag 1 and apply the F -test for Granger non-causality (Granger, 1969). We report the asymmetry between alignment $\rightarrow$ performance and performance $\rightarrow$ alignment significance (Table 6). The asymmetric pattern holds across six Pythia scales (70M–2.8B) and three of four MixNet variants.

Table 6. **Bidirectional Granger causality test results at lag 1.** Bold p values indicate $p < 0.05$.

Model	align $\rightarrow$ perf (p)	perf $\rightarrow$ align (p)
Pythia-70M	1.1e-3	1.000
Pythia-160M	6.8e-4	1.000
Pythia-410M	4.1e-5	1.000
Pythia-1B	1.4e-3	0.043
Pythia-1.4B	2.5e-5	1.000
Pythia-2.8B	1.2e-5	1.000
MixNet-S	0.045	0.277
MixNet-M	0.025	0.022
MixNet-L	0.026	0.014
MixNet-XL	0.370	0.442

Partial correlation controlling for model scale. We partial $\log_{10}$ parameter count out of the performance–alignment relationship using the standard partial correlation formula. The relationship remains significant in both modalities (Table 7).

Table 7. **Partial Pearson correlation between performance and brain alignment after controlling for $\log_{10}$ parameter count.**

Modality	Benchmark	r_{xy}	$r_{x,\log p}$	$r_{y,\log p}$	$r_{\text{partial}} (p)$
Language (n=35)	Average	0.880	0.885	0.839	0.544 ($p = 8.8 \times 10^{-4}$)
	BBH	0.827	0.885	0.833	0.351 ($p = 0.042$)
	MMLU-PRO	0.812	0.885	0.839	0.273 ($p = 0.119$)
Vision (n=578)	ImageNet	0.496	0.413	0.760	0.307 ($p = 5.1 \times 10^{-14}$)
	ImageNet-A	0.400	0.413	0.808	0.124 ($p = 2.9 \times 10^{-3}$)
	ImageNet-R	0.441	0.413	0.794	0.204 ($p = 7.8 \times 10^{-7}$)
	sketch	0.447	0.413	0.804	0.212 ($p = 2.7 \times 10^{-7}$)

Noise ceiling. For each of the 180 HCP regions we compute the inter-subject CKA across the four NSD subjects (using pairwise mean), yielding a per-region noise ceiling. Across all language models, mean alignment meets or exceeds the ceiling in 78% of regions; for vision models, 53%. Stratifying by benchmark tertile (Table 8) reveals a clean monotonic gradient that is particularly pronounced for language models.

Table 8. **Ceiling-reaching rate (% of 180 HCP regions where mean alignment $\geq$ inter-subject ceiling) stratified by model performance tier.**

Modality	Bottom 1/3	Middle 1/3	Top 1/3
Language	6%	47%	88%
Vision	36%	58%	62%

Wasserstein distance between vision and language manifolds. For each of the 8 normalized depth bins, we compute the 1-Wasserstein distance between the distribution of vision-model alignment vectors and the distribution of language-model alignment vectors (Table 9). The distance decreases monotonically with depth, supporting the qualitative claim (Section 3.2) that deeper vision layers approach the language-model manifold.

Table 9. Vision–language alignment manifold distance across normalized depth bins.

Depth bin	Wasserstein ↓	Cosine sim ↑	Euclidean ↓
1/8	0.01519	0.9827	0.2433
2/8	0.01371	0.9876	0.2210
3/8	0.01232	0.9891	0.2009
4/8	0.01073	0.9914	0.1772
5/8	0.00946	0.9922	0.1588
6/8	0.00796	0.9938	0.1361
7/8	0.00609	0.9954	0.1079
8/8	0.00564	0.9955	0.1016
Spearman ρ	−1.00	+1.00	−1.00

Layer-wise depth trend statistics. Following the reviewer’s suggestion, we provide explicit statistical tests for the layer-depth alignment trend (Figure 3a). Linear regression of mean alignment on the 8 normalized depth bins yields $R^2 = 0.984$ ($p = 1.3 \times 10^{-6}$, Spearman $\rho = 0.976$) for the 594 vision models, and $R^2 = 0.628$ ($p = 0.011$, Spearman $\rho = 0.850$) for the 38 language models, consistent with the visually apparent monotonic rise in vision and the middle-layer plateau in language.



Figure 7. Taxonomic organization of AI models based on brain alignment patterns. Radial dendrogram showing hierarchical clustering of models based on their brain alignment vectors (180 brain regions $\times$ 8 normalized depth layers $\times$ 4 subjects). Each node represents a model, with node color indicating architecture class (CNN, Transformer, Hybrid), shape indicating modality (circles for language models, stars for vision models), and size representing the alignment score.